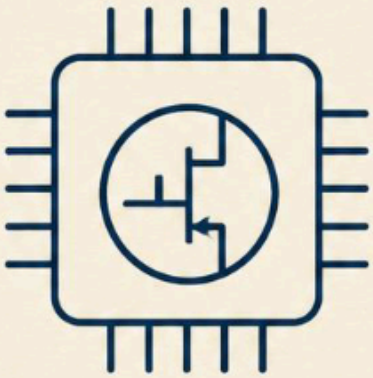
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ELECTRONIC DEVICES & CIRCUITS

PART 2

FETs • AMPLIFIERS • FEEDBACK • OSCILLATORS



CONCEPTS • CIRCUITS • APPLICATIONS

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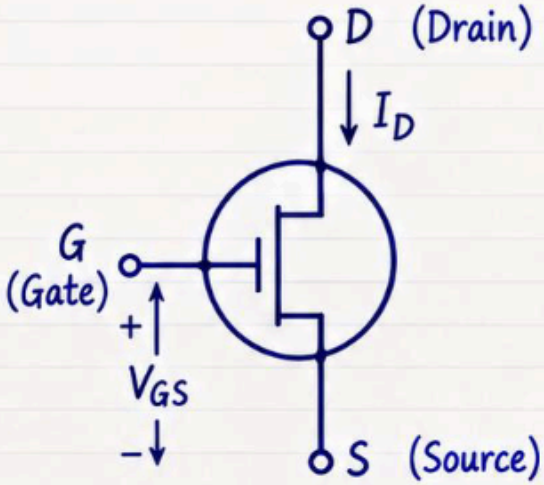
FIELD-EFFECT TRANSISTORS

What is a FET?

① A FET is a three-terminal semiconductor device.

② Drain current is controlled by an electric field produced by gate voltage.

③ It is a voltage-controlled, unipolar device with very high input impedance.

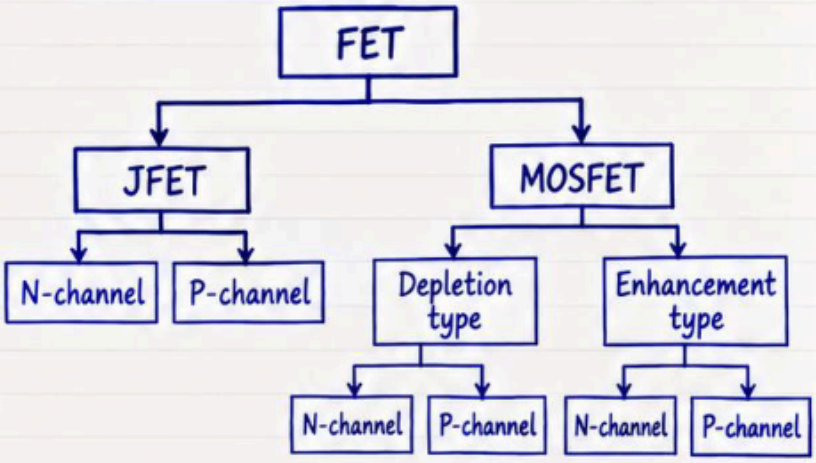


V_{GS} controls I_D

Main Features

- Majority-carrier operation.
- High input impedance.
- Low noise.
- Low power consumption.
- Good thermal stability.

FET Family



Common Applications

- Amplifier.
- Analog switch.
- Buffer (voltage follower).
- Voltage-controlled resistor.
- CMOS / digital circuits.

Remember:

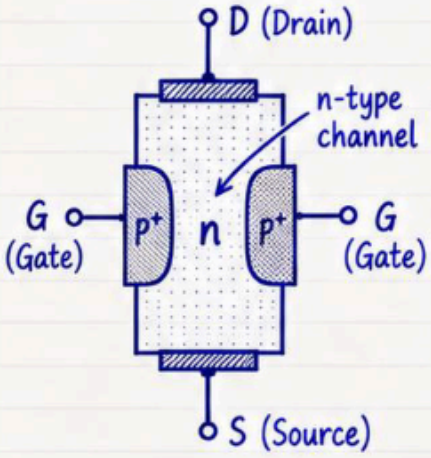
BJT is current-controlled.
FET is voltage-controlled.

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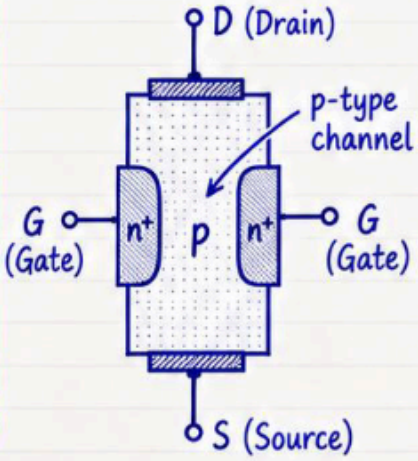
JFET CONSTRUCTION & WORKING

Construction

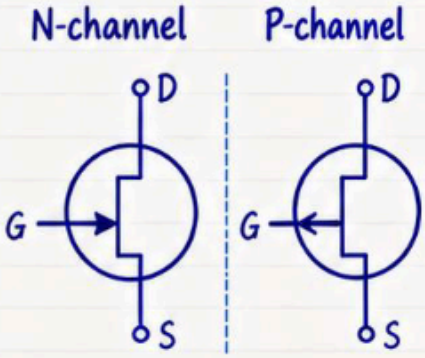
N-channel JFET



P-channel JFET



Circuit Symbols



Arrow points into the channel for N-channel and out of the channel for P-channel.

Working of N-Channel JFET

① **$V_{GS} = 0$ and small V_{DS} :**
Channel offers low resistance and current I_D flows through the channel (from drain to source).

② **Increasing V_{DS} :**
Depletion regions near the drain widen, so the channel becomes narrower towards the drain end.

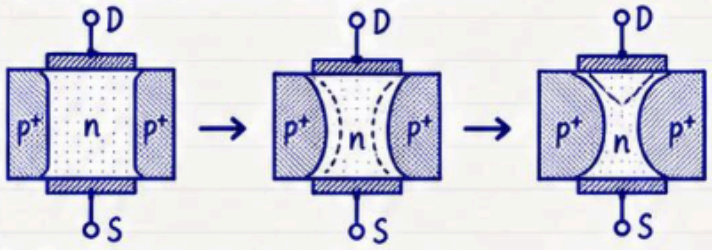
③ **At $V_{DS} \approx |V_P|$:**
Depletion regions just meet at the drain end. The channel is pinched-off and current saturates.

④ **Beyond pinch-off :**
I_D remains nearly constant at I_{DSS} (saturation region) until breakdown occurs at a very high V_{DS}.

Effect of Gate Voltage

- For N-channel JFET, V_{GS} is negative (w.r.t. source).
- Negative V_{GS} widens the depletion regions, narrows the channel and reduces the drain current I_D.
- At $V_{GS} = V_{GS(off)} = -|V_P|$, the channel is completely cut off and $I_D \approx 0$.

Channel Condition (N-Channel)



Wide channel ($V_{GS} = 0$, small V_{DS})

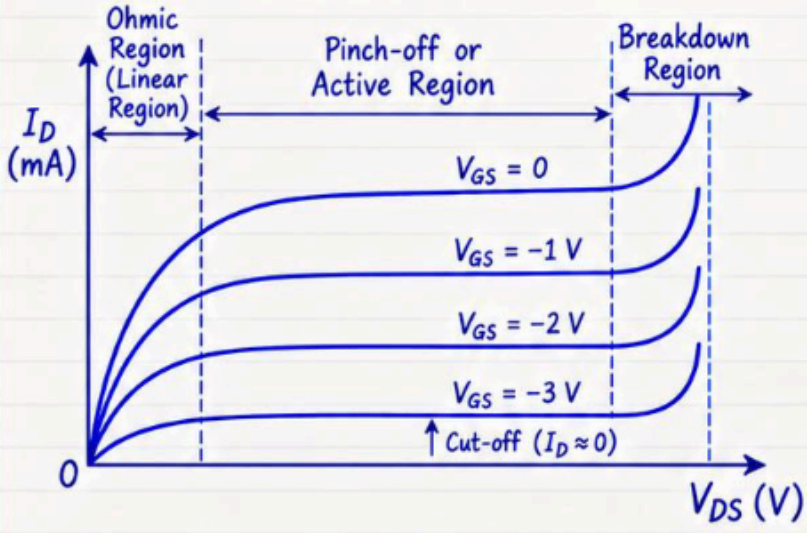
Channel narrowing (increasing V_{DS})

Pinch-off ($V_{DS} \approx |V_P|$)

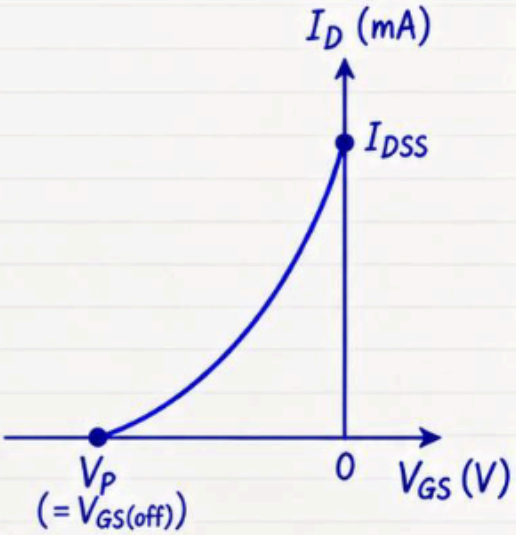
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JFET CHARACTERISTICS

1. Drain Characteristics



2. Transfer Characteristic



3. Shockley Equation

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$V_P \le V_{GS} \le 0$$

I_D = drain current (A)
I_{DSS} = drain current at $V_{GS} = 0$ (A) (maximum drain current)
V_P = pinch-off voltage (V) ($V_{GS(off)}$), negative for n-channel
V_{GS} = gate-to-source voltage (V)

4. Important Parameters

Transconductance (g_m)

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}}$$

(S or mS)

At $V_{GS} = 0$, $g_{m0} = \frac{2I_{DSS}}{|V_P|}$

Drain resistance (r_d)

$$r_d = \frac{\Delta V_{DS}}{\Delta I_D}$$

(Ω or kΩ)

Amplification factor (μ)

$$\mu = g_m r_d$$

(dimensionless)

5. Worked Example

Given: $I_{DSS} = 12$ mA, $V_P = -6$ V, $V_{GS} = -2$ V
Find: I_D
Solution:

$$I_D = 12\left(1 - \frac{(-2)}{(-6)}\right)^2 = 12\left(\frac{2}{3}\right)^2 = 12 \times \frac{4}{9} = 5.33 \text{ mA}$$

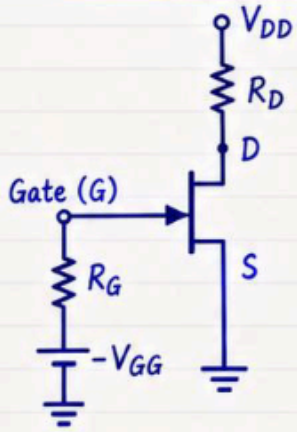
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FET BIASING & COMPARISON

Why Biasing?

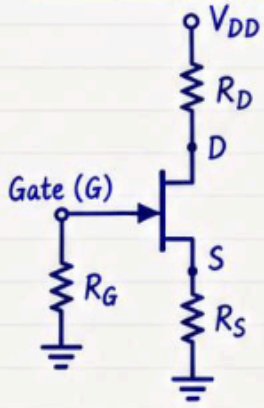
① Biasing fixes a stable Q-point in the active region.
② It permits undistorted amplification.

1. Fixed-bias JFET



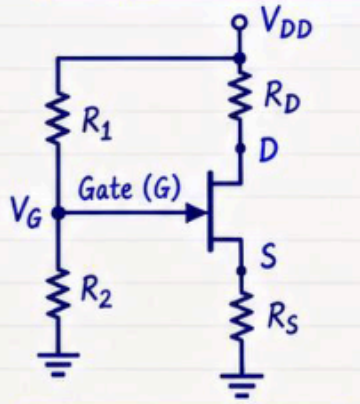
$$V_{GS} = -V_{GG}$$

2. Self-bias JFET



$$V_{GS} = -I_D R_S$$

3. Voltage-divider bias



$$V_G = V_{DD} \frac{R_2}{R_1 + R_2}$$

$$V_{GS} = V_G - I_D R_S$$

Q-point Relation (Shockley Equation)

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

I_D : drain current
I_{DSS} : drain current at $V_{GS} = 0$
V_P : pinch-off voltage (V_P is negative for n-channel JFET)

BJT vs JFET vs MOSFET

Parameter	BJT	JFET	MOSFET
Control	Current controlled	Voltage controlled	Voltage controlled
Carriers	Bipolar (electrons & holes)	Unipolar (majority carriers)	Unipolar (majority carriers)
Input impedance	Low - medium	High	Very high
Noise	Moderate	Low	Low
Typical use	High-gain stages	Low-noise input	Switching and CMOS

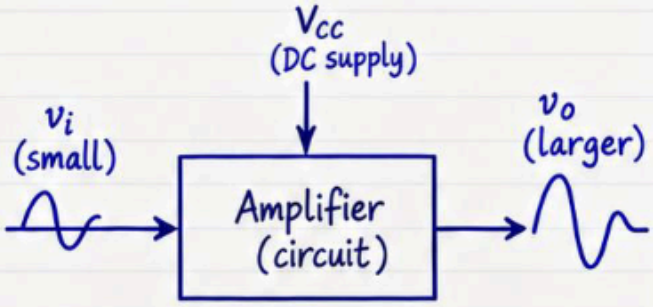
Key Point: FET gate current is ideally almost zero.

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SMALL-SIGNAL AMPLIFIERS

1. Purpose

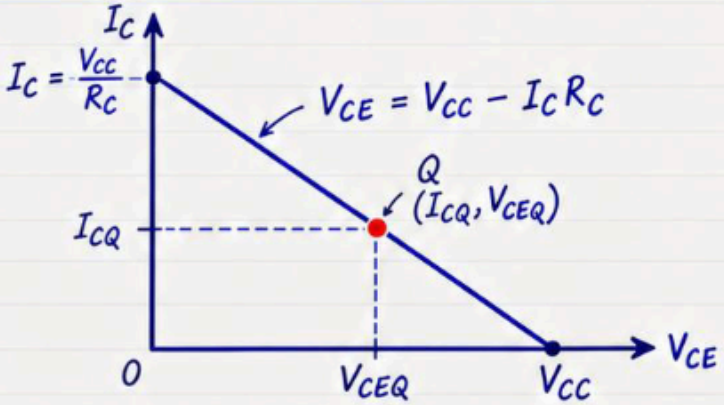
- An amplifier uses DC supply power to produce a larger output replica of a small AC input signal.
- Proper biasing places the device at a Q-point so that the output can swing without distortion.

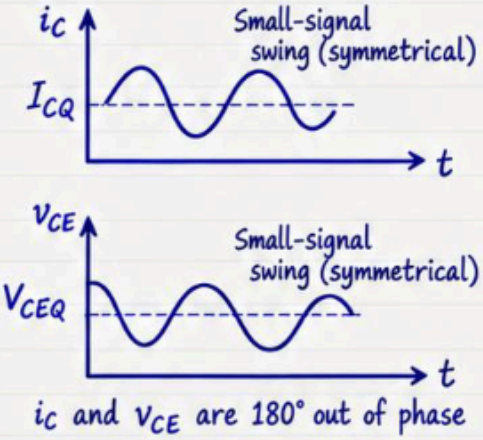


2. Important Gains

Voltage gain	Current gain	Power gain	Gain in decibels (for voltage gain)
$A_v = \frac{v_o}{v_i}$	$A_i = \frac{i_o}{i_i}$	$A_p = \frac{P_o}{P_i} \approx A_v A_i$	$A_{v(dB)} = 20 \log_{10} \lvert A_v \rvert$

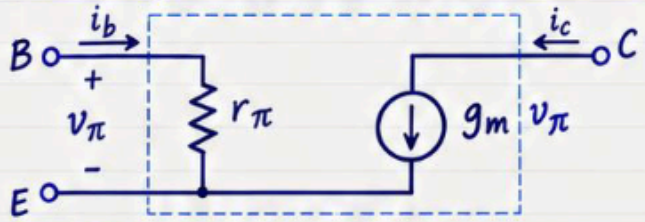
3. DC Load Line & Q-Point





i_C and v_{CE} are 180° out of phase

4. Small-Signal BJT Model (hybrid-π)



$$g_m = \frac{I_C}{V_T}$$

$$r_\pi = \frac{\beta}{g_m}$$

$V_T \approx 25\ mV$ (at room temperature)

5. Good Amplifier Needs

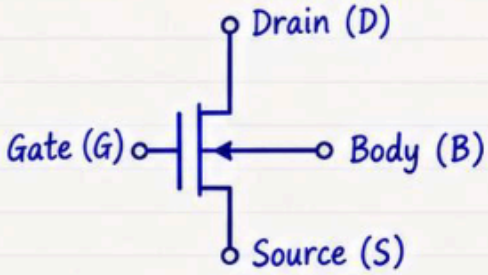
- Sufficient gain
- Suitable input and output impedance
- Wide bandwidth
- Low distortion

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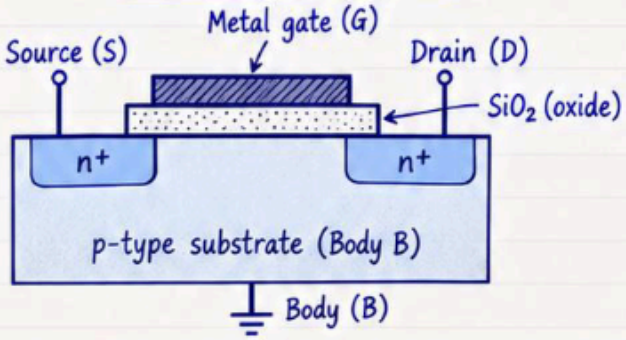
MOSFET FUNDAMENTALS

What is a MOSFET?

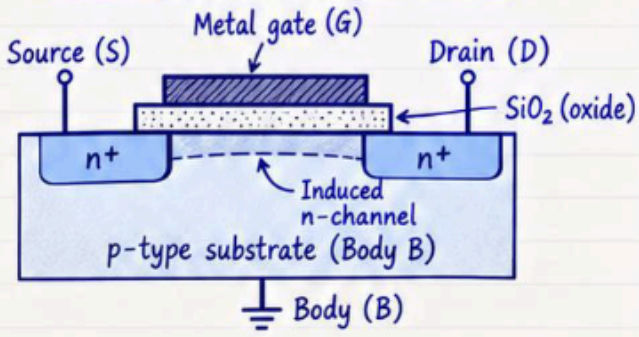
A Metal-Oxide-Semiconductor FET (MOSFET) has an insulated gate separated from the channel by a thin layer of SiO_2, giving extremely high input impedance.



$V_{GS} = 0$ (No channel)

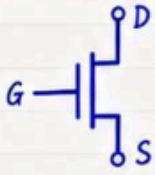


$V_{GS} > V_{TH}$ (Induced n-channel)



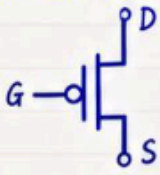
Types of MOSFETs

Enhancement NMOS



Normally OFF (channel induced for $V_{GS} > V_{TH}$)

Enhancement PMOS



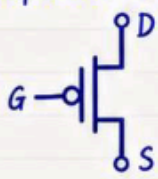
Normally OFF (channel induced for $V_{SG} > |V_{TH}|$)

Depletion NMOS



Normally ON (conducts at $V_{GS} = 0$)

Depletion PMOS



Normally ON (conducts at $V_{SG} = 0$)

Operating Regions – Enhancement NMOS

Region	Condition	Drain Current (I_D)
Cut-off	$V_{GS} < V_{TH}$	$I_D \approx 0$
Triode (Linear)	$V_{GS} > V_{TH}$ $0 \le V_{DS} < V_{GS} - V_{TH}$	I_D increases (with V_{DS})
Saturation	$V_{GS} > V_{TH}$ $V_{DS} \ge V_{GS} - V_{TH}$	$I_D = k(V_{GS} - V_{TH})^2$ (ideal square-law)

Overdrive voltage:

$$V_{OV} = V_{GS} - V_{TH}$$

and in saturation,

$$I_D = k(V_{GS} - V_{TH})^2$$

Applications

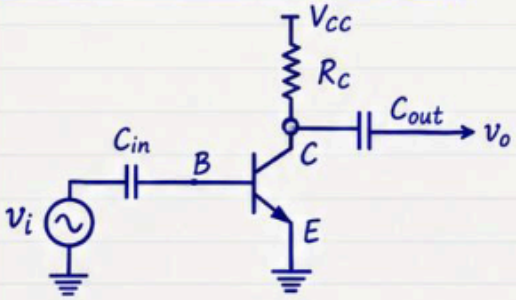
- CMOS logic
- Switching
- Amplifiers
- Power control

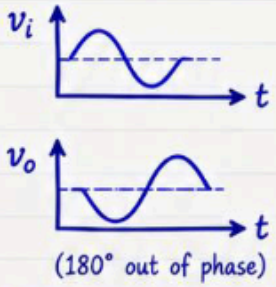
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BJT AMPLIFIER CONFIGURATIONS

- BJT amplifiers are classified by the terminal common to input and output.

1. Common Emitter (CE)

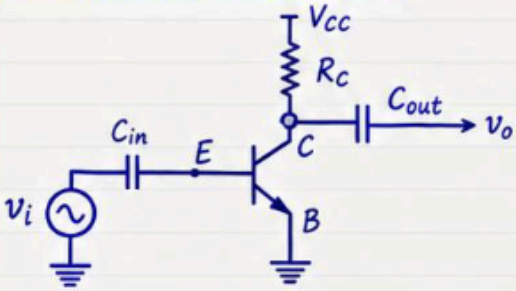


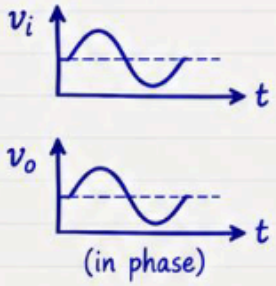


(180° out of phase)

- Input at base, output at collector.
- Emitter is AC-grounded.
- Voltage gain: high
- Current gain: high
- Input impedance: medium
- Output impedance: high
- Output is 180° out of phase with input.

2. Common Base (CB)

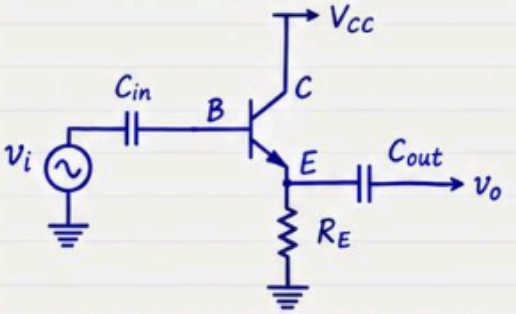


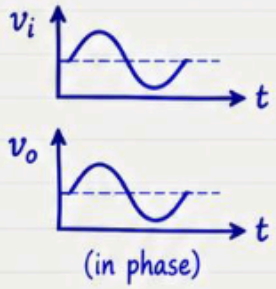


(in phase)

- Input at emitter, output at collector.
- Base is AC-grounded.
- Voltage gain: high
- Current gain: less than 1
- Input impedance: low
- Output impedance: high
- No phase reversal (output in phase with input).

3. Common Collector (CC) / Emitter Follower





(in phase)

- Input at base, output at emitter.
- Collector is connected to V_{CC} (AC ground).
- Voltage gain: approximately 1
- Current gain: high
- Input impedance: high
- Output impedance: low
- No phase reversal (output in phase with input).

4. Comparison Table

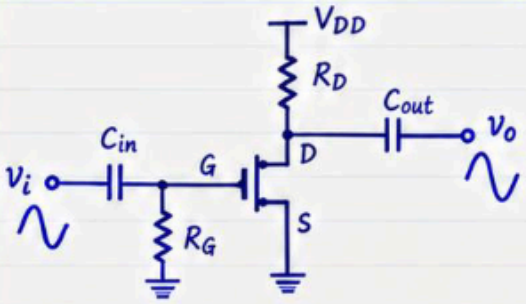
Parameter	Common Emitter (CE)	Common Base (CB)	Common Collector (CC)
Voltage gain	High	High	≈ 1 (slightly less)
Current gain	High	Less than 1	High
Input impedance	Medium	Low	High
Output impedance	High	High	Low
Phase shift (V_o w.r.t. V_i)	180° (inverted)	0° (in phase)	0° (in phase)
Common use	General amplification	High-frequency stage	Buffer / Impedance matching

Most widely used: Common Emitter

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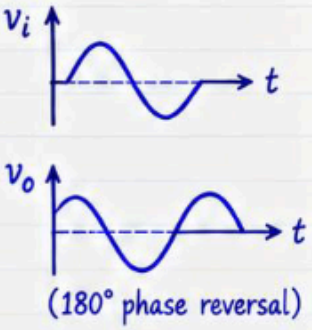
FET AMPLIFIERS

1. Common Source (CS) Amplifier

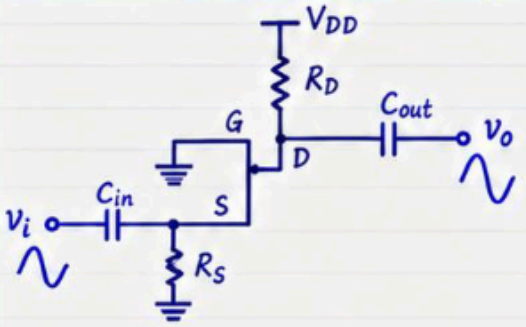


- Source is common (AC grounded).
- Input at gate, output at drain.
- High voltage gain.
- Very high input impedance (≈ open circuit).
- Output is 180° out of phase with input.

$$A_v \approx -g_m(R_D \parallel R_L)$$

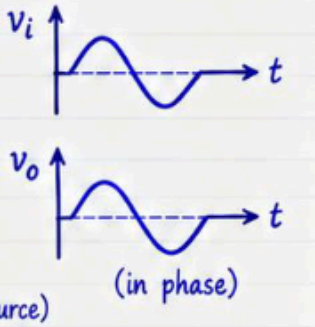


2. Common Gate (CG) Amplifier

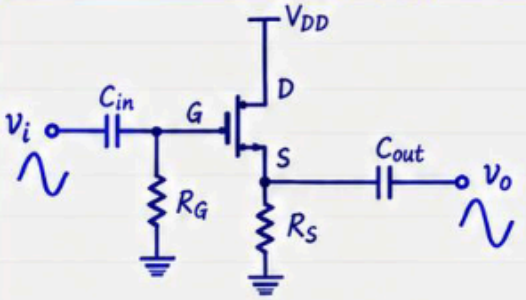


- Gate is common (AC grounded).
- Input at source, output at drain.
- High voltage gain.
- Low input impedance.
- High output impedance.
- No phase reversal (output in phase with input).

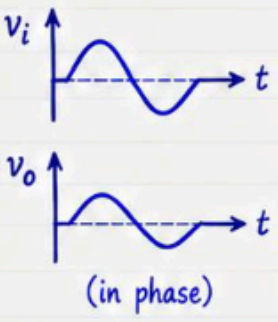
$$R_{in} \approx \frac{1}{g_m} \quad \text{(at input, looking into source)}$$



3. Common Drain (CD) / Source Follower



- Drain is common (connected to V_{DD}).
- Input at gate, output at source.
- Voltage gain slightly less than 1 ($A_v < 1$).
- Very high input impedance.
- Low output impedance.
- No phase reversal (output in phase with input).



4. Comparison of FET Amplifier Configurations

Parameter	Common Source (CS)	Common Gate (CG)	Common Drain (CD)
Voltage gain (A_v)	High $A_v \approx -g_m(R_D \parallel R_L)$	High	Slightly less than 1 ($A_v < 1$)
Input impedance (R_{in})	Very high (≈ ∞)	Low (≈ $1/g_m$)	Very high (≈ ∞)
Output impedance (R_{out})	High	High	Low
Phase shift	180° (inverted)	0° (in phase)	0° (in phase)
Application	Voltage amplifier	High-frequency / current buffer	Impedance matching (buffer)

5. Remember

FET configuration is analogous to BJT configuration:

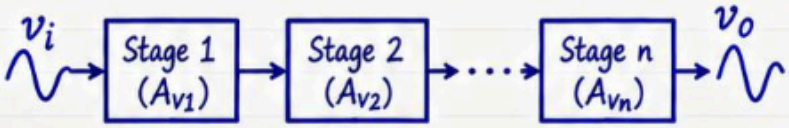
CS ↔ CE
CG ↔ CB
CD ↔ CC

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MULTISTAGE & RC-COUPLED AMPLIFIERS

1. Why Multistage?

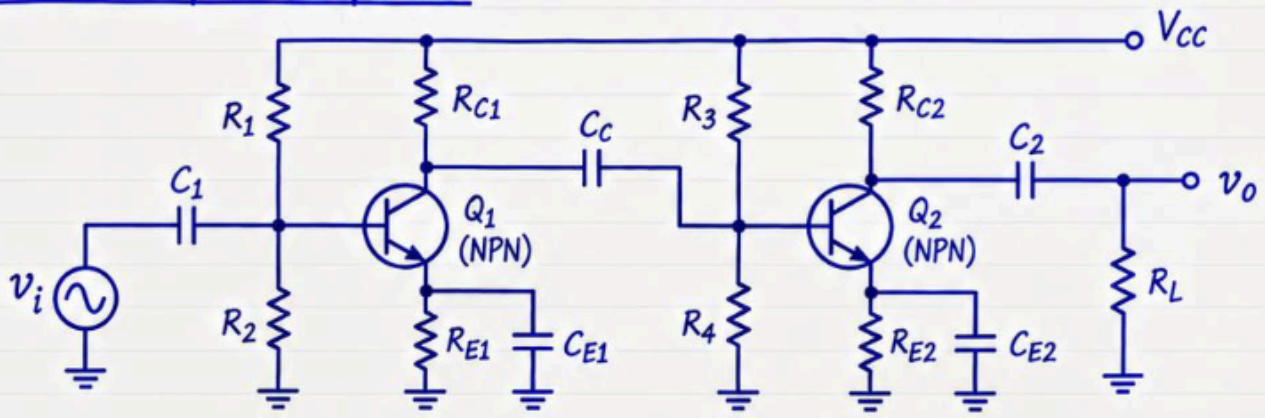
- A single amplifier stage may not provide enough voltage gain to meet the required level.
- Cascading multiple stages increases the total gain.



$$A_{v(total)} = A_{v1} \times A_{v2} \times \cdots \times A_{vn}$$

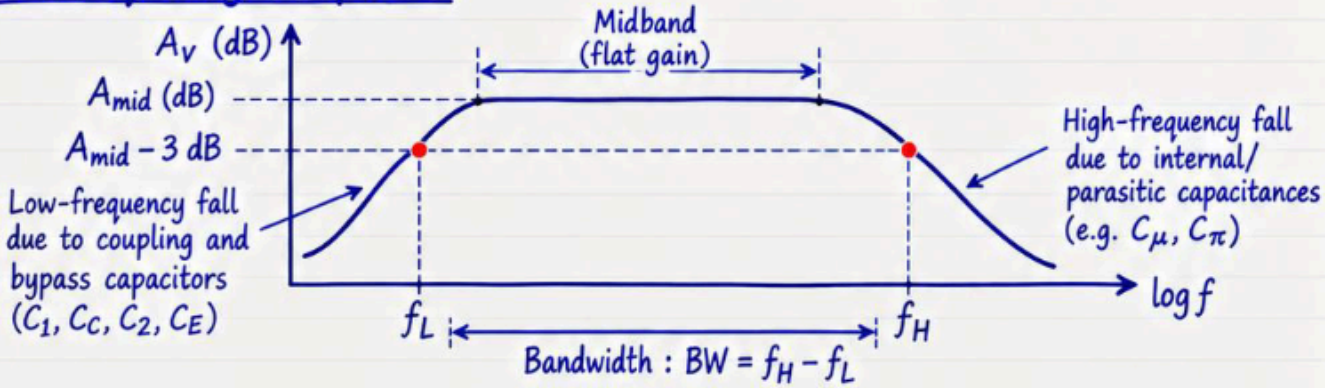
$$\text{Gain in dB} = A_{v1(dB)} + A_{v2(dB)} + \cdots + A_{vn(dB)}$$

2. RC-Coupled Amplifier



- Coupling capacitors (C_1, C_C, C_2) pass AC signals but block DC, so each stage maintains its own DC bias (operating point) independently.

3. Frequency Response



4. Example

- If a two-stage amplifier has $A_{v1} = 20$ and $A_{v2} = 30$,

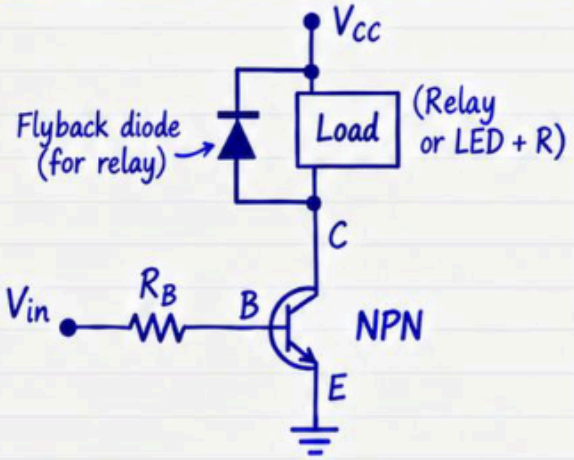
$$A_{v(total)} = 20 \times 30 = 600$$

$$\text{Gain in dB} = 20 \log_{10}(600) \approx 55.6 \text{ dB}$$

@crafted_note

SWITCHING CIRCUITS & MULTIVIBRATORS

1. Transistor as a Switch



Two States of Operation

Condition	Key values	Load status
OFF (cut-off)	$I_B = 0$ $I_C \approx 0$ $V_{CE} \approx V_{CC}$	Load OFF (no current)
ON (saturation)	I_B (sufficient) $V_{CE(sat)} \approx 0.2\ V$ I_C = load current	Load ON (current flows)

Base resistor selection:

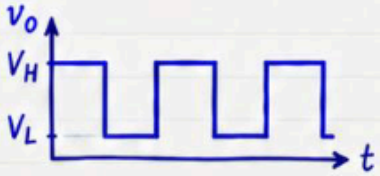
$$R_B = \frac{V_{in} - V_{BE}}{I_B}$$

$V_{BE} \approx 0.7\ V$ (for silicon)

2. Multivibrators

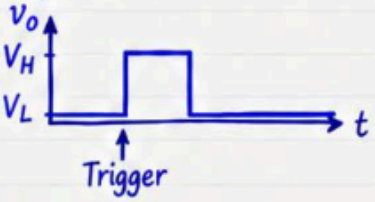
(a) Astable Multivibrator

- No stable state
- Free-running oscillator
- Produces continuous square wave output



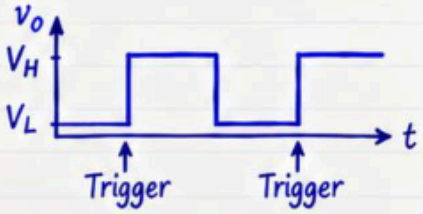
(b) Monostable Multivibrator

- One stable state
- Produces one pulse per trigger (one-shot)
- Used as a timer / pulse generator



(c) Bistable Multivibrator

- Two stable states
- Changes state on each trigger
- Used as a flip-flop / memory element



3. Quick Revision

JFET (Shockley equation)	MOSFET (saturation region)	Amplifier gain	Negative feedback	Oscillation (Barkhausen criterion)
$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$	$I_D = k\,(V_{GS} - V_{TH})^2$	$A_v = \frac{v_o}{v_i}$	$A_f = \frac{A}{1 + A\beta}$	$\lvert A\beta \rvert = 1$ phase = 0° or 360°

4. Common Mistakes

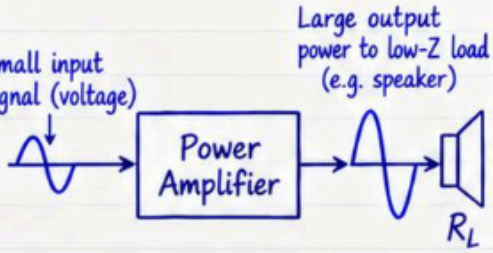
- Do not omit the flyback diode across a relay coil.
- Do not confuse cut-off with saturation.
- Check device polarity and connections (NPN vs PNP, diode direction, supply polarity).

@crafted_note

POWER AMPLIFIERS

1. Purpose

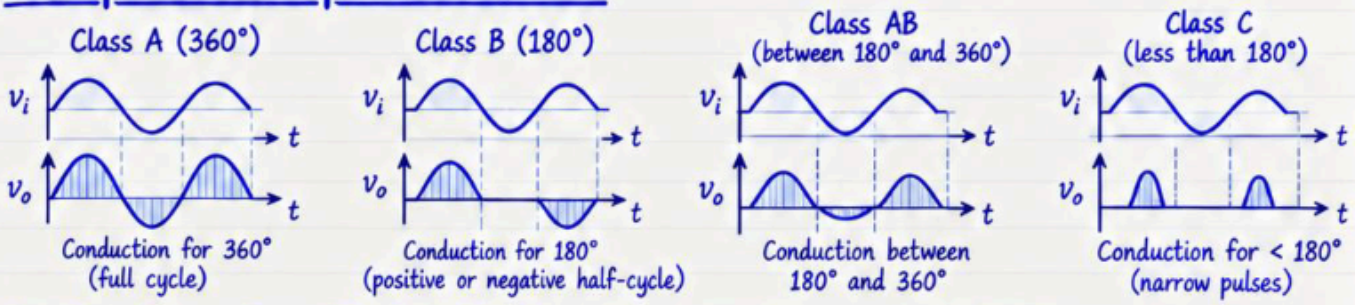
- Power amplifiers deliver large AC power to a low-impedance load such as a speaker.
- They are judged by output power, efficiency, heat dissipation and distortion.



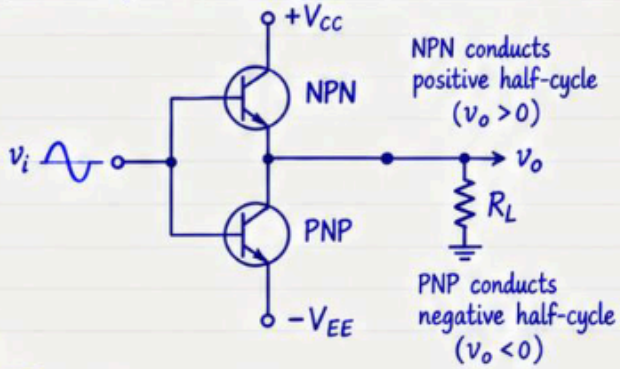
2. Amplifier Classes

Class	Conduction angle (θ)	Q-point / Biasing	Key features	Maximum efficiency (ideal)	Typical use
A	360°	Q-point at middle of load line	• Best linearity • Low efficiency	25% (resistive load) 50% (transformer coupling)	High fidelity audio
B	180°	Biased at cut-off (Q-point at cut-off)	• Higher efficiency • Crossover distortion	78.5% (theoretical)	Audio power amplifiers
AB	Between 180° and 360°	Biased slightly above cut-off	• Lower crossover distortion than Class B • Efficiency between Class A and B	Between 25% and 78.5%	Audio power amplifiers
C	Less than 180°	Biased well below cut-off	• High efficiency • High distortion	High (can be > 80%)	Tuned RF circuits

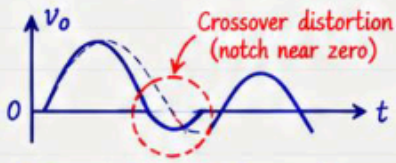
3. Input and Output Waveforms



4. Complementary Push-Pull Class B



5. Crossover Distortion



Class AB reduces crossover distortion by providing a slight forward bias to both transistors.

6. Important Formulas

Power amplifier efficiency	Load power (rms)
$\eta = \frac{P_{AC(out)}}{P_{DC(in)}} \times 100\%$	$P_L = \frac{V_{L(rms)}^2}{R_L}$